Mechanism bridge — activation chemistry from the reaction, not the family name

Deterministic classifier reads the native reaction's products; fixes the family-keyword mis-calls.

products contain NMP + diphosphate

e.g. ...+ ATP = ...+ AMP + diphosphate

→ acyl-adenylate

→ attack ATP Pα (O3A leaving axis)

products contain NDP + phosphate

e.g. ...+ ATP = ...+ ADP + phosphate

→ acyl-phosphate

→ attack ATP Pγ (O3B leaving axis)

Validation against the structural pipeline's prior (family-keyword) assignment

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agree where reaction data exists

1

corrected (ddaG: family said none; reaction → Pα)

9

no reaction annotation → honest 'unresolved' (was silently defaulted)

ddaG (E2JA32) — the case that proves the point

InterPro family “Fumarate–diaminopropanoate ligase” carries no adenylation hint.

Its reaction (... = ... + AMP + diphosphate) unambiguously → Pα. The old keyword map scored it on Pγ

(0 productive poses → 'inconclusive'); the reaction-derived call fixes it. Now emitted with full provenance.